Department of Computer Engineering

Faculty of Engineering, Kasetsart University

HW#2: Relational Database and Relational Algebra

Part 1: Relational Algebra

Consider the following database schema:

Employee(Fname, Lname, Ssn, Address, Salary, Dno, SuperSsn)

Department(Dname, Dnumber, MgrSsn, MgrStartDate)

Project(Pname, Pnumber, Plocation, Dnum)

Works_on(Essn, Pno, Hours)

Dependent(Essn, DependentName, Sex, Bdate, Relationship)

Write a Relational Algebra expression for each query below. You may use the Relational Algebra operators introduced in class, such as:

Operator	Meaning
σ	Selection
π	Projection
$\times$	Cartesian Product
$\bowtie, \bowtie_{condition}$	Natural-Join, Teta-Join
$\cup$	Union
$\cap$	Intersection
$-$	Difference
ρ	Rename

1.1 Find the first and last names of employees who live in Bangkok.

1.2 Find the first and last names of employees whose salary is greater than 50,000 Baht.

1.3 Find the names of departments that control at least one project located in Bangkok.

1.4 Find the names of employees who work more than 20 hours per week on a project.

1.5 Find the names of employees who are department managers and also have at least one dependent.

1.6 Find the names of projects in which no employee from the “Research” department participates.

Requirements

- Write the complete Relational Algebra expression for each query.
- Use meaningful names when applying the Rename operator.
- The final result should contain only the attributes requested in the query.

Part 2: Database Design and Application

Choose ONE of the following business domains:

1. Hospital Management System — examples: Doctors, Patients, Appointments, Examination Rooms
2. Hotel Management System — examples: Rooms, Reservations, Customers, Employees
3. Airline Management System — examples: Flights, Passengers, Aircraft, Airports
4. School Management System — examples: Students, Teachers, Courses, Classrooms

You may introduce additional entities if necessary.

2.1 Database Schema Design

Design a relational database for your selected domain. Your database must contain:

- 3–5 relations/tables
- At least 5 attributes per table
- Appropriate data types
- Clearly identified Primary Keys
- Clearly identified Foreign Keys

For each table, present the design in a format similar to:

Attribute	Data Type	Key	Description
StudentID	VARCHAR(10)	PK	Unique student identifier
StudentName	VARCHAR(100)		Student's full name
Email	VARCHAR(100)		Student email address
Major	VARCHAR(50)		Student's major
BirthDate	DATE		Student's date of birth

Also explain the relationships among the tables using either a diagram or a short textual explanation.

Student 1 ——————< Enrollment >—————— 1 Course

Key Analysis

For each table, identify:

- At least 2 Super Keys
- All reasonable Candidate Keys
- The selected Primary Key
- Foreign Key(s), if any

Briefly explain why you selected the Primary Key. For each Foreign Key, clearly indicate the reference.

Example: Enrollment.StudentID ⟶ Student.StudentID

2.2 Sample Data

Create sample data for your database.

- At least 5 records per table
- The data must be realistic and internally consistent.
- Primary Key values must be unique.
- Foreign Key values must correctly reference existing records.
- The sample data should support the business questions in Part 2.3.
- All sample data must be created by you.

2.3 Business Questions and Relational Algebra

Create five meaningful business questions based on your database. Each question must demonstrate the specified Relational Algebra concept.

Question 1 — Selection + Projection

Create one business question requiring Selection (**σ**) and Projection (**π**). Write the Relational Algebra expression.

Question 2 — Natural Join

Create one business question requiring a Natural Join. Write the Relational Algebra expression.

Question 3 — Theta Join

Create one business question requiring a Theta Join. Clearly specify the join condition and write the Relational Algebra expression.

Question 4 — Set Operation

Create one business question requiring Union, Intersection, or Difference. Write the Relational Algebra expression and make sure that the relations involved are union-compatible.

Question 5 — Multiple Operations

Create one more complex business question requiring at least 2–3 Relational Algebra operations (for example, Selection ⟶ Join ⟶ Projection). Write the complete Relational Algebra expression.

***** Extra Credit *****

Create **ONE additional challenging business question** that requires a Relational Algebra operation or concept **not covered in detail in class**.

Possible examples include:

- Division
- Aggregation
- COUNT
- MIN
- MAX
- AVG
- GROUP BY

Submission

Submit ONE PDF file containing all answers.

Filename: StudentID_HW#2.pdf

Your PDF should contain:

1. Part 1 — Relational Algebra (Questions 1.1–1.6)
2. Part 2 — Database Design and Application (2.1 Database Schema Design and Key Analysis, 2.2 Sample Data, and 2.3 Five Business Questions and Relational Algebra) + Extra (Optional !)

Important Rules

1. Your work must be your own.

Copying answers, database designs, sample data, or Relational Algebra expressions from another student or another source may result in zero credit.

2. AI Usage Policy

Do not use AI tools to generate answers for this assignment.

AI tools may be used only for grammar checking, spelling correction, and improving the English of text you have already written.

AI tools must not be used to generate database schemas, key analysis, sample data, business questions, or Relational Algebra expressions.

The purpose of this homework is to practice the fundamental database skills yourself before using AI-assisted tools.